004_strategy

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1 004_strategy

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1.1 Purpose

This stage converts structured event-response data into **evaluated trading candidates**.

Rather than defining a deployable trading system, this stage focuses on: - measuring directional consistency of cross-asset reactions - identifying repeatable lead–lag behavior - constructing candidate signals for evaluation

The goal is to determine whether observed relationships represent: - noise - weak statistical signal - or viable strategy candidates

1.2 Inputs

This notebook consumes artifacts from:

- output/002_enrich/
- output/003_analysis/

Specifically:

- `reaction_panel.pkl` — event-level return responses across assets
- `shock_events.pkl` — identified shock events from analysis stage
- `sigma_event_matrix.pkl` — event structure and alignment signals
- `event_bitmap.pkl` *(diagnostic)* — cross-asset activation structure

1.3 Analysis Goals

This stage performs four key tasks:

- 1. Construct Event Response Dataset Build a unified dataset describing how assets respond following a reference asset shock.
- 2. Measure Directional Consistency Evaluate whether follower assets move in the expected direction relative to the initiating event.
- 3. Identify Aligned Events Filter events where direction and magnitude meet consistency criteria.
- 4. Aggregate Pair-Level Statistics Compute hit rates and frequencies for asset pairs to assess signal strength.

1.4 Outputs

Artifacts generated by this stage:

- `event_response_df.pkl` — base event response dataset
- `event_response_expanded.pkl` — enriched dataset with derived features
- `trade_df.pkl` — mapped trade representation of signals
- `daily_returns.pkl` — aggregated return series
- `reduced_df.pkl` — filtered dataset of aligned events
- `pair_hit.pkl` — pair-level hit rate statistics

These artifacts are written to:

- output/004_strategy/

and registered in the pipeline `manifest.pkl`.

1.5 Pipeline Position

The pipeline follows the following steps:

- 001_download –> data acquisition
- 002_enrich –> transformation and enrichment
- 003_analysis –> statistical structure
- **004_strategy –> exploration and hypothesis evaluation**
- 005_backtest –> execution mapping and performance evaluation

1.6 Design Principle

This stage follows these principles:

- **Deterministic pipeline** — no recomputation of prior stages
- **No forward assumptions** — all results derived from observed data
- **Separation from backtesting** — evaluation only, not deployment
- **Full artifact persistence** — every transformation is saved

This stage intentionally stops short of claiming a tradable strategy, and instead produces **evidence for or against a statistical edge**.

1.6.1 1. Imports and Environment Setup

1.6.2 Provide the necessary imports required for to to proceed.

[253]:
```
import pandas as pd
import numpy as np
import pickle
from datetime import datetime, UTC
```

```python
import math
from pathlib import Path
import matplotlib.pyplot as plt
```

1.6.3 2. Prepare the environment for the notebook

[254]:
```python
startdate = "2023-01-01"
trading_days = 252
frequency = "1d"

universe = [
    "BTCUSDT",   # Bitcoin
    "ETHUSDT",   # Ethereum
    "BNBUSDT",   # Binance Coin
    "SOLUSDT",   # Solana
    "XRPUSDT",   # Ripple
    "ADAUSDT",   # Cardano
    "DOGEUSDT",  # Dogecoin
    "AVAXUSDT",  # Avalanche
    "LTCUSDT"    # Litecoin
]

execution_delay = [0, 1, 2, 3]
execution_cost_bps = [20, 30, 40]

stage_label = "004_strategy"

OUTPUT_ROOT = Path("../output")
OUTPUT_ROOT.mkdir(parents=True, exist_ok=True)

MANIFEST_FILE = OUTPUT_ROOT / "manifest.pkl"

DOWNLOAD_DIR = OUTPUT_ROOT / "001_download"
DOWNLOAD_DIR.mkdir(parents=True, exist_ok=True)

ENRICH_DIR = OUTPUT_ROOT / "002_enrich"
ENRICH_DIR.mkdir(parents=True, exist_ok=True)

ANALYSIS_DIR = OUTPUT_ROOT / "003_analysis"
ANALYSIS_DIR.mkdir(parents=True, exist_ok=True)

STRATEGY_DIR = OUTPUT_ROOT / "004_strategy"
STRATEGY_DIR.mkdir(parents=True, exist_ok=True)

inspection_window = 20

sigma_threshold = 3
```

```
observation_window_length = 10
observation_window = range(1, observation_window_length + 1)

holding_period = 1
```

1.6.4 3. Loading the manifest pickle file

[255]:
```
if MANIFEST_FILE.exists():
    manifest = pd.read_pickle(MANIFEST_FILE)
else:
    manifest = {}

manifest.setdefault(stage_label, {})
```

[255]: {}

1.6.5 4. Load the pickle files from the previous stage

We use these file contents for our analysis.

[256]:
```
# 001_download artifacts
prices = pd.read_pickle(DOWNLOAD_DIR / "PRICES.pkl")
event_panels = {}
for coin in universe:
    panel_file = DOWNLOAD_DIR / f"{coin}.event_panel.pkl"
    event_panels[coin] = pd.read_pickle(panel_file)

# 002_enrich artifacts
returns_full = pd.read_pickle(ENRICH_DIR / "returns_full.pkl")
z_scores = pd.read_pickle(ENRICH_DIR / "z_scores.pkl")
rolling_sigma = pd.read_pickle(ENRICH_DIR / "rolling_sigma.pkl")
price_wide = pd.read_pickle(ENRICH_DIR / "price_wide.pkl")

# 003_analysis artifacts
event_bitmap = pd.read_pickle(ANALYSIS_DIR / "event_bitmap.pkl")
extreme_z_scores = pd.read_pickle(ANALYSIS_DIR / "extreme_z_scores.pkl")
sigma_event_matrix = pd.read_pickle(ANALYSIS_DIR / "sigma_event_matrix.pkl")

print("prices:", prices.shape)
print("returns_full:", returns_full.shape)
print("z_scores:", z_scores.shape)
print("rolling_sigma:", rolling_sigma.shape)
print("price_wide:", price_wide.shape)
print("event_bitmap:", event_bitmap.shape)
print("extreme_z_scores:", extreme_z_scores.shape)
print("sigma_event_matrix:", sigma_event_matrix.shape)
```

```
print("event panels loaded:", list(event_panels.keys()))
```

```
prices: (10408, 5)
returns_full: (983, 9)
z_scores: (964, 9)
rolling_sigma: (964, 9)
price_wide: (1178, 9)
event_bitmap: (19, 11)
extreme_z_scores: (9, 5)
sigma_event_matrix: (9, 5)
event panels loaded: ['BTCUSDT', 'ETHUSDT', 'BNBUSDT', 'SOLUSDT', 'XRPUSDT',
'ADAUSDT', 'DOGEUSDT', 'AVAXUSDT', 'LTCUSDT']
```

1.6.6 5. Reconstruct sigma_class from sigma_event_matrix

[259]:
```
assert "sigma_event_matrix" in locals(), "sigma_event_matrix not loaded"

sigma_class = (z_scores > sigma_threshold).astype(int) - (z_scores <␣
↪-sigma_threshold).astype(int)

print("max:", z_scores.max().max())
print("min:", z_scores.min().min())
print("event_count:", ((z_scores >= 3) | (z_scores <= -3)).sum().sum())

sigma_class = (z_scores >= sigma_threshold).astype(int) - (z_scores <=␣
↪-sigma_threshold).astype(int)

assert sigma_class.shape == z_scores.shape
assert int((sigma_class != 0).sum().sum()) == 104

print("OK:", sigma_class.shape, "events:", int((sigma_class != 0).sum().sum()))
```

```
max: 4.281477768984525
min: -4.155916934268655
event_count: 104
OK: (964, 9) events: 104
```

1.6.7 6. Review Shock Events

[260]:
```
sigma_class = np.floor(z_scores).astype(int)

# count signed sigma values
sigma_counts = sigma_class.stack().value_counts().sort_index()

print(sigma_counts)
```

```
-5       4
-4      40
-3     184
```

```
-2     892
-1    3221
 0    3061
 1     946
 2     268
 3      58
 4       2
Name: count, dtype: int64
```

1.6.8 7. compute a mask to filter out middle values

[261]:
```
extreme_mask = (sigma_coin_counts.index <= -3) | (sigma_coin_counts.index >= 3)
```

1.6.9 8. Explore breakdown of events

[262]:
```
# count by signed sigma and coin
pd.set_option("display.width", 2000)
pd.set_option("display.max_columns", None)
stacked = sigma_class.stack()

sigma_coin_counts = (
    stacked
    .groupby([stacked, stacked.index.get_level_values(1)])
    .size()
    .unstack(fill_value=0)
    .sort_index()
)

print(sigma_coin_counts)
```

```
coin  ADAUSDT  AVAXUSDT  BNBUSDT  BTCUSDT  DOGEUSDT  ETHUSDT  LTCUSDT  SOLUSDT
XRPUSDT
-5          0         0        0        1         0        1        0        0
2
-4          6         3        5        3         3        5        7        3
5
-3         16        19       27       24        21       22       19       18
18
-2        111       109       89       91        91      102       96      108
95
-1        367       356      332      355       383      340      357      353
378
 0        325       335      367      341       329      355      356      315
338
 1        103       106      102      111       101      106       99      129
89
 2         33        31       35       31        28       26       22       33
29
```

```
 3          2          5          7          7          8          6          8          5
10
 4          1          0          0          0          0          1          0          0
0
```

1.6.10 9. apply the mask to get number of extreme events per coin

[263]:
```python
extreme_sigma_coin_counts = sigma_coin_counts.loc[extreme_mask]

print(extreme_sigma_coin_counts)
```

```
coin  ADAUSDT  AVAXUSDT  BNBUSDT  BTCUSDT  DOGEUSDT  ETHUSDT  LTCUSDT  SOLUSDT
XRPUSDT
-5          0         0        0        1         0        1        0        0
2
-4          6         3        5        3         3        5        7        3
5
-3         16        19       27       24        21       22       19       18
18
 3          2         5        7        7         8        6        8        5
10
 4          1         0        0        0         0        1        0        0
0
```

1.6.11 10. reorganize sigma, coin, and date

[264]:
```python
leader_mask = (sigma_class >= 3) | (sigma_class <= -3)
leader_events = sigma_class.where(leader_mask)

# convert to long format
event_df = (
    leader_events
    .stack()
    .reset_index()
)

# standardize column names
event_df.columns = ["event_date", "reference_coin", "sigma"]

event_df["sigma"] = event_df["sigma"].astype(int)

# sort for sanity
event_df = event_df.sort_values(["event_date", "reference_coin"]).
 ↪reset_index(drop=True)

# quick checks
print("event_df shape:", event_df.shape)
print(event_df.head())
```

```
event_df shape: (288, 3)
                 event_date reference_coin  sigma
0 2023-08-03 00:00:00+00:00        LTCUSDT     -3
1 2023-08-15 00:00:00+00:00       AVAXUSDT     -3
2 2023-08-15 00:00:00+00:00       DOGEUSDT     -3
3 2023-08-15 00:00:00+00:00        SOLUSDT     -3
4 2023-08-15 00:00:00+00:00        XRPUSDT     -3
```

1.6.12 11. Attach forward return vectors

[265]:
```
all_coins = price_wide.columns.tolist()

records = []

for _, row in event_df.iterrows():
    event_date = row["event_date"]
    ref_coin = row["reference_coin"]
    sigma = row["sigma"]

    # locate event index
    if event_date not in price_wide.index:
        continue

    idx = price_wide.index.get_loc(event_date)

    for target_coin in all_coins:
        if target_coin == ref_coin:
            continue

        # ensure full forward window exists
        if idx + observation_window_length >= len(price_wide):
            continue

        # extract forward prices
        price_series = price_wide[target_coin].iloc[idx : idx +␣
↪observation_window_length + 1]

        # convert to returns
        returns = price_series.pct_change().iloc[1:].values

        if len(returns) != observation_window_length:
            continue

        records.append({
            "event_date": event_date,
            "reference_coin": ref_coin,
            "target_coin": target_coin,
```

```
            "sigma": sigma,
            "returns_vector": returns
        })

event_response_df = pd.DataFrame(records)

print("event_response_df shape:", event_response_df.shape)
print(event_response_df.head())

EVENT_RESPONSE_RAW_FILE = STRATEGY_DIR / "event_response_df.pkl"
event_response_df.to_pickle(EVENT_RESPONSE_RAW_FILE)
manifest[stage_label]["event_response_df"] = str(EVENT_RESPONSE_RAW_FILE)
print("saved:", EVENT_RESPONSE_RAW_FILE)
print("shape:", event_response_df.shape)
```

```
event_response_df shape: (2304, 5)
                 event_date reference_coin target_coin  sigma
returns_vector
0 2023-08-03 00:00:00+00:00        LTCUSDT     ADAUSDT     -3
[0.004098360655737654, -0.0010204081632652073,…
1 2023-08-03 00:00:00+00:00        LTCUSDT    AVAXUSDT     -3
[-0.003212851405622441, 0.00241740531829171, 0…
2 2023-08-03 00:00:00+00:00        LTCUSDT     BNBUSDT     -3
[0.0016625103906899863, 0.007883817427385864, …
3 2023-08-03 00:00:00+00:00        LTCUSDT     BTCUSDT     -3
[-0.00256893391640145, -0.0018550830546302244,…
4 2023-08-03 00:00:00+00:00        LTCUSDT    DOGEUSDT     -3
[-0.0017643865363733413, 0.028687967369136702,…
saved: ../output/004_strategy/event_response_df.pkl
shape: (2304, 5)
```

1.6.13 12. attach the values for returns into a vector we can use

[266]:
```
lags = 10

# expand returns_vector into columns
returns_expanded = pd.DataFrame(
    event_response_df["returns_vector"].tolist(),
    columns=[f"lag_{i+1}" for i in range(lags)]
)

# combine back
event_response_expanded = pd.concat(
    [event_response_df.drop(columns=["returns_vector"]), returns_expanded],
    axis=1
)

print(event_response_expanded.shape)
```

```
print(event_response_expanded.head())

EVENT_RESPONSE_EXPANDED_FILE = STRATEGY_DIR / "event_response_expanded.pkl"
event_response_expanded.to_pickle(EVENT_RESPONSE_EXPANDED_FILE)
manifest[stage_label]["event_response_expanded"] =␣
 ↪str(EVENT_RESPONSE_EXPANDED_FILE)
print("saved:", EVENT_RESPONSE_EXPANDED_FILE)
print("shape:", event_response_expanded.shape)
```

```
(2304, 14)
                 event_date reference_coin target_coin  sigma     lag_1
lag_2     lag_3     lag_4     lag_5     lag_6     lag_7     lag_8     lag_9
lag_10
0 2023-08-03 00:00:00+00:00        LTCUSDT     ADAUSDT     -3  0.004098
-0.001020 -0.006129 -0.003768  0.023384  0.011089 -0.014955 -0.009447 -0.006131
-0.006511
1 2023-08-03 00:00:00+00:00        LTCUSDT    AVAXUSDT     -3 -0.003213
0.002417  0.011254 -0.011924  0.021722 -0.004724 -0.011867 -0.004804 -0.004023
-0.008078
2 2023-08-03 00:00:00+00:00        LTCUSDT     BNBUSDT     -3  0.001663
0.007884  0.000412 -0.004938  0.014475 -0.006115 -0.010254 -0.005387  0.000417
0.000416
3 2023-08-03 00:00:00+00:00        LTCUSDT     BTCUSDT     -3 -0.002569
-0.001855  0.000888  0.003721  0.019690 -0.006716 -0.004086 -0.000918 -0.000073
-0.003871
4 2023-08-03 00:00:00+00:00        LTCUSDT    DOGEUSDT     -3 -0.001764
0.028688 -0.017182 -0.011565  0.020952  0.004797  0.005570 -0.001846  0.013742
-0.024374
saved: ../output/004_strategy/event_response_expanded.pkl
shape: (2304, 14)
```

1.6.14 13. Mean response by sigma and lag

[267]:
```
lag_cols = [f"lag_{i+1}" for i in range(10)]

sigma_response = (
    event_response_expanded
    .groupby("sigma")[lag_cols]
    .mean()
    .sort_index()
)

print(sigma_response)
```

```
          lag_1     lag_2     lag_3     lag_4     lag_5     lag_6     lag_7
lag_8     lag_9    lag_10
sigma
-5     0.031235  0.005290  0.004586 -0.014319 -0.018990  0.012551 -0.006291
```

```
0.006079  0.009411 -0.002635
-4     0.023121  0.015307  0.008495 -0.006337 -0.010505  0.022606 -0.015286
0.013943  0.011418 -0.008052
-3     0.000979  0.004238 -0.000655  0.002586 -0.004354  0.013425 -0.007493
0.007244 -0.002948  0.000122
 3     0.001176  0.015192 -0.001047  0.001384  0.005148 -0.004364 -0.004639
-0.003317  0.009706  0.005610
 4    -0.043455  0.012152  0.020587 -0.029966 -0.013297  0.039237 -0.058328
-0.025557  0.024736  0.023787
```

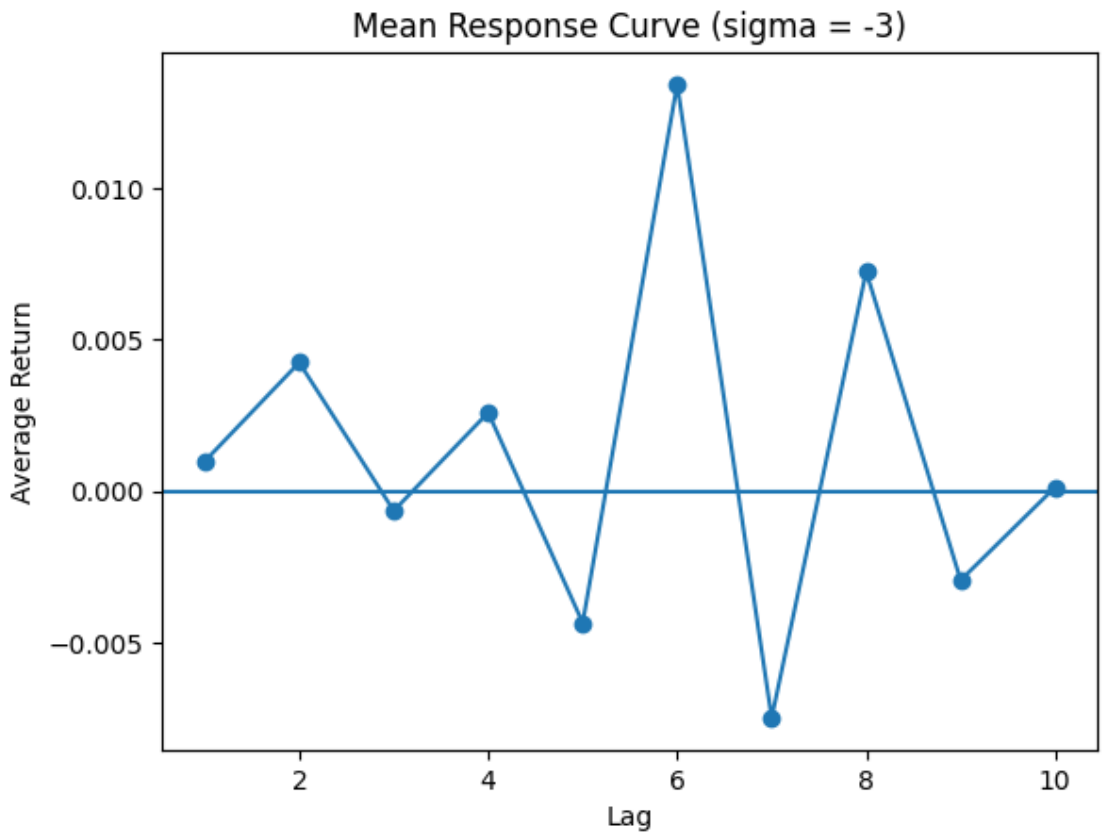
1.6.15 14. Exploratory: Focus on sigma = -3 only

[268]:
```
sigma3 = event_response_expanded[
    event_response_expanded["sigma"] == -3
]

sigma3_mean = sigma3[lag_cols].mean()

print(sigma3_mean)
```

```
lag_1     0.000979
lag_2     0.004238
lag_3    -0.000655
lag_4     0.002586
lag_5    -0.004354
lag_6     0.013425
lag_7    -0.007493
lag_8     0.007244
lag_9    -0.002948
lag_10    0.000122
dtype: float64
```

1.6.16 15. Plot the results for the lags to see if any patterns.

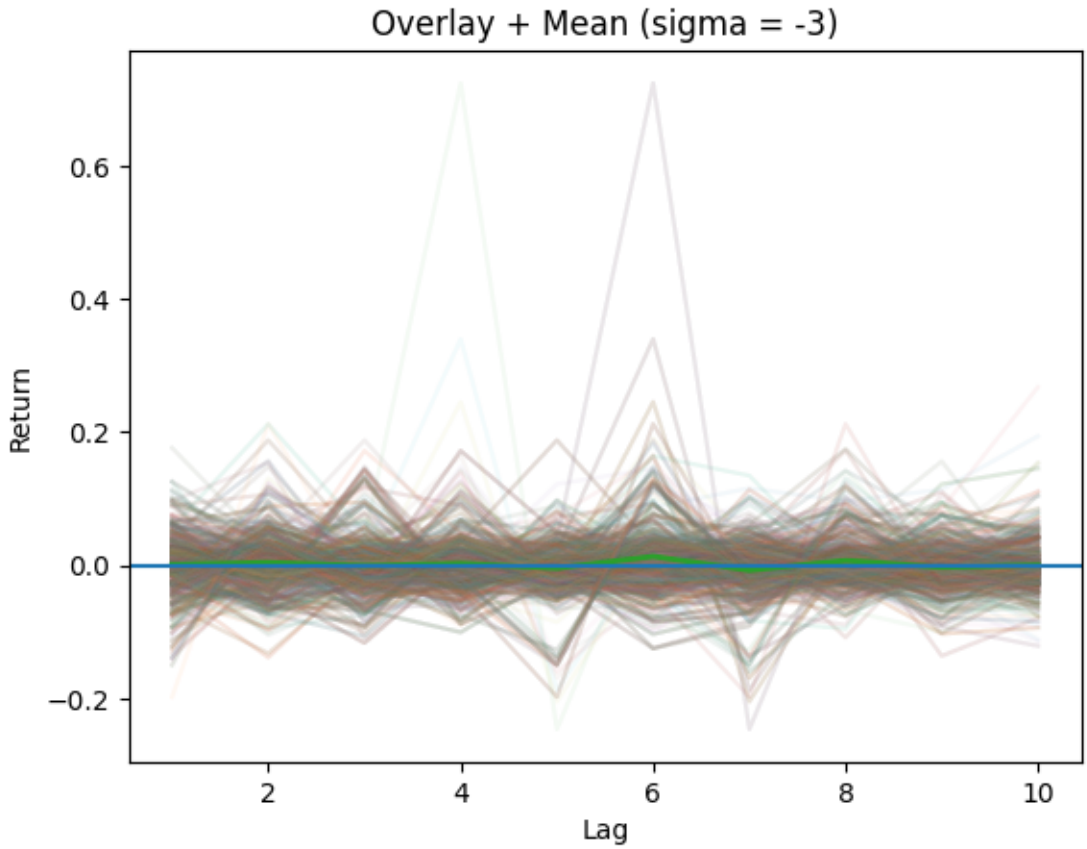
[269]:
```
lags = list(range(1, 11))
values = sigma3_mean.values

plt.figure()
plt.plot(lags, values, marker='o')
plt.axhline(0)

plt.title("Mean Response Curve (sigma = -3)")
plt.xlabel("Lag")
plt.ylabel("Average Return")

plt.show()
```



1.6.17 16. Plot all events, to see the vectors

[270]:
```
lags = list(range(1, 11))

# use RAW dataframe (has returns_vector)
sigma3_raw = event_response_df[
    event_response_df["sigma"] == -3
]

plt.figure()

# overlay all paths
for vec in sigma3_raw["returns_vector"]:
    plt.plot(lags, vec, alpha=0.05)

# mean line (from expanded dataframe)
plt.plot(lags, sigma3_mean.values, linewidth=2)

plt.axhline(0)
```

```
plt.title("Overlay + Mean (sigma = -3)")
plt.xlabel("Lag")
plt.ylabel("Return")

plt.show()
```



1.6.18 17. Dispersion analysis per lag (sigma = -3)

[271]:
```
lag_cols = [f"lag_{i+1}" for i in range(10)]

dispersion_df = pd.DataFrame({
    "mean": sigma3[lag_cols].mean(),
    "std": sigma3[lag_cols].std(),
})

# signal-to-noise proxy
dispersion_df["abs_mean"] = dispersion_df["mean"].abs()
dispersion_df["snr"] = dispersion_df["abs_mean"] / dispersion_df["std"]
```

```
print(dispersion_df.sort_index())
```

```
            mean       std  abs_mean       snr
lag_1   0.000979  0.047237  0.000979  0.020729
lag_10  0.000122  0.036284  0.000122  0.003364
lag_2   0.004238  0.045786  0.004238  0.092555
lag_3  -0.000655  0.045908  0.000655  0.014267
lag_4   0.002586  0.046073  0.002586  0.056134
lag_5  -0.004354  0.041042  0.004354  0.106086
lag_6   0.013425  0.066320  0.013425  0.202434
lag_7  -0.007493  0.045628  0.007493  0.164225
lag_8   0.007244  0.038653  0.007244  0.187411
lag_9  -0.002948  0.036422  0.002948  0.080928
```

1.6.19 18. Lag_5 to Lag_6 transition test

[272]:
```
subset = sigma3.copy()

# condition: lag_5 negative
subset = subset[subset["lag_5"] < 0]

print("count:", len(subset))

print("\nlag_6 stats (given lag_5 < 0):")
print("mean:", subset["lag_6"].mean())
print("median:", subset["lag_6"].median())
print("positive_ratio:", (subset["lag_6"] > 0).mean())
```

```
count: 817

lag_6 stats (given lag_5 < 0):
mean: 0.014590768089947309
median: 0.009396902310288002
positive_ratio: 0.598531211750306
```

1.6.20 19. Refined test: sigma = -3 + lag_5 filters

[273]:
```
# base filter: sigma = -3 only
subset = event_response_expanded[
    event_response_expanded["sigma"] == -3
].copy()

print("base count (sigma = -3):", len(subset))

### condition 1: lag_5 < 0
subset_weak = subset[subset["lag_5"] < 0]

print("\n--- Condition: lag_5 < 0 ---")
```

```
print("count:", len(subset_weak))
print("mean:", subset_weak["lag_6"].mean())
print("median:", subset_weak["lag_6"].median())
print("positive_ratio:", (subset_weak["lag_6"] > 0).mean())

### condition 2: lag_5 < -1% (stronger signal)
subset_strong = subset[subset["lag_5"] < -0.01]

print("\n--- Condition: lag_5 < -1% ---")
print("count:", len(subset_strong))
print("mean:", subset_strong["lag_6"].mean())
print("median:", subset_strong["lag_6"].median())
print("positive_ratio:", (subset_strong["lag_6"] > 0).mean())
```

```
base count (sigma = -3): 1472

--- Condition: lag_5 < 0 ---
count: 817
mean: 0.014590768089947309
median: 0.009396902310288002
positive_ratio: 0.598531211750306

--- Condition: lag_5 < -1% ---
count: 547
mean: 0.017958162340694507
median: 0.011550678602368025
positive_ratio: 0.5923217550274223
```

1.6.21 20. Conditioned leader –> follower test

[275]:
```
### build base dataset
base = event_response_expanded.copy()

print("total rows:", len(base))

### condition filter (leader condition)
conditioned = base[
    (base["sigma"] == -3) &
    (base["lag_5"] < -0.01)
].copy()

print("\nconditioned rows:", len(conditioned))

### baseline (for comparison)
baseline = base[
    base["sigma"] == -3
].copy()
```

```python
print("baseline rows:", len(baseline))

### function to compute stats
def summarize(df, label):
    print(f"\n--- {label} ---")

    for lag in ["lag_1", "lag_2", "lag_3"]:
        mean = df[lag].mean()
        median = df[lag].median()
        pos = (df[lag] > 0).mean()

        print(f"{lag}:")
        print(f"  mean:   {mean:.5f}")
        print(f"  median: {median:.5f}")
        print(f"  pos%:   {pos:.3f}")

### run comparison
summarize(baseline, "BASELINE (sigma = -3)")
summarize(conditioned, "CONDITIONED (sigma=-3 & lag_5<-1%)")
```

```
total rows: 2304

conditioned rows: 547
baseline rows: 1472

--- BASELINE (sigma = -3) ---
lag_1:
  mean:   0.00098
  median: 0.00225
  pos%:   0.531
lag_2:
  mean:   0.00424
  median: 0.00498
  pos%:   0.537
lag_3:
  mean:   -0.00065
  median: -0.00344
  pos%:   0.461

--- CONDITIONED (sigma=-3 & lag_5<-1%) ---
lag_1:
  mean:   -0.00966
  median: -0.00790
  pos%:   0.426
lag_2:
  mean:   0.01309
  median: 0.01458
  pos%:   0.645
```

```
lag_3:
  mean:   -0.00188
  median: 0.00053
  pos%:   0.519
```

1.6.22 21. FINAL SIGNAL VALIDATION (lag_2)

[276]:
```python
subset = event_response_expanded[
    (event_response_expanded["sigma"] == -3) &
    (event_response_expanded["lag_5"] < -0.01)
].copy()

lag = subset["lag_2"]

print("===== SAMPLE =====")
print("count:", len(lag))

print("\n===== CORE STATS =====")
print("mean:", lag.mean())
print("median:", lag.median())
print("std:", lag.std())

print("\n===== WIN / LOSS =====")
win_rate = (lag > 0).mean()
avg_win = lag[lag > 0].mean()
avg_loss = lag[lag <= 0].mean()

print("win_rate:", win_rate)
print("avg_win:", avg_win)
print("avg_loss:", avg_loss)
print("win/loss ratio:", avg_win / abs(avg_loss) if avg_loss != 0 else None)

print("\n===== TAIL RISK =====")
q = lag.quantile([0.01, 0.05, 0.1, 0.9, 0.95, 0.99])
print(q)

print("\n===== EXPECTED VALUE CHECK =====")
expected_value = win_rate * avg_win + (1 - win_rate) * avg_loss
print("expected_value:", expected_value)
```

```
===== SAMPLE =====
count: 547

===== CORE STATS =====
mean: 0.013085454873295698
median: 0.014584154513204384
std: 0.04282153500298594
```

```
===== WIN / LOSS =====
win_rate: 0.6453382084095064
avg_win: 0.0366040122959591
avg_loss: -0.029708621261756794
win/loss ratio: 1.2321006745297394

===== TAIL RISK =====
0.01   -0.096806
0.05   -0.065324
0.10   -0.041753
0.90    0.064272
0.95    0.088804
0.99    0.118180
Name: lag_2, dtype: float64

===== EXPECTED VALUE CHECK =====
expected_value: 0.013085454873295696
```

1.6.23 22. Trade mapping (conditional timing rule)

[277]:
```
df = event_response_expanded.copy()

### signal condition
df["use_lag2"] = (
    (df["sigma"] == -3) &
    (df["lag_5"] < -0.01)
)

### choose return based on rule
df["signal_return"] = 0.0

# conditioned → lag_2
df.loc[df["use_lag2"], "signal_return"] = df.loc[df["use_lag2"], "lag_2"]

# default → lag_1 (only for sigma = -3 events)
default_mask = (df["sigma"] == -3) & (~df["use_lag2"])
df.loc[default_mask, "signal_return"] = df.loc[default_mask, "lag_1"]

### keep only actual trades
trade_df = df[df["signal_return"] != 0].copy()

print("total trades:", len(trade_df))
print("lag2 trades:", trade_df["use_lag2"].sum())
print("lag1 trades:", (~trade_df["use_lag2"]).sum())

TRADE_FILE = STRATEGY_DIR / "trade_df.pkl"
trade_df.to_pickle(TRADE_FILE)
```

```
manifest[stage_label]["trade_df"] = str(TRADE_FILE)
print("saved:", TRADE_FILE)
print("shape:", trade_df.shape)
```

```
total trades: 1470
lag2 trades: 547
lag1 trades: 923
saved: ../output/004_strategy/trade_df.pkl
shape: (1470, 16)
```

1.6.24 23. Build PnL time series

[278]:
```
### aggregate by date (equal weight)
daily_returns = (
    trade_df
    .groupby("event_date")["signal_return"]
    .mean()
    .sort_index()
)

print("days:", len(daily_returns))
print("mean daily return:", daily_returns.mean())
print("std daily return:", daily_returns.std())

### equity curve
equity_curve = (1 + daily_returns).cumprod()

### basic stats
total_return = equity_curve.iloc[-1] - 1
volatility = daily_returns.std() * np.sqrt(trading_days)

sharpe = (
    daily_returns.mean() /
    daily_returns.std()
) * np.sqrt(trading_days)

drawdown = equity_curve / equity_curve.cummax() - 1
max_drawdown = drawdown.min()

print("\n===== PERFORMANCE =====")
print("total_return:", total_return)
print("volatility:", volatility)
print("sharpe:", sharpe)
print("max_drawdown:", max_drawdown)

DAILY_RETURNS_FILE = STRATEGY_DIR / "daily_returns.pkl"
daily_returns.to_pickle(DAILY_RETURNS_FILE)
manifest[stage_label]["daily_returns"] = str(DAILY_RETURNS_FILE)
```

```
print("saved:", DAILY_RETURNS_FILE)
print("shape:", daily_returns.shape)
```

```
days: 64
mean daily return: 0.003958412022438095
std daily return: 0.040329976387437544

===== PERFORMANCE =====
total_return: 0.22202289504911277
volatility: 0.6402185274136012
sharpe: 1.5580927245018186
max_drawdown: -0.20238612360361818
saved: ../output/004_strategy/daily_returns.pkl
shape: (64,)
```

1.6.25 24. Apply transaction costs

[279]:
```
cost_levels = {
    "20bps": 0.002,
    "30bps": 0.003,
    "40bps": 0.004,
}

results = {}

for label, cost in cost_levels.items():

    df_cost = trade_df.copy()

    # subtract cost per trade
    df_cost["net_return"] = df_cost["signal_return"] - cost

    # aggregate daily
    daily = (
        df_cost
        .groupby("event_date")["net_return"]
        .mean()
        .sort_index()
    )

    # equity
    equity = (1 + daily).cumprod()

    # stats
    total_return = equity.iloc[-1] - 1
    vol = daily.std() * np.sqrt(trading_days)
```

```
    sharpe = (
        daily.mean() /
        daily.std()
    ) * np.sqrt(trading_days)

    dd = equity / equity.cummax() - 1
    max_dd = dd.min()

    results[label] = {
        "total_return": total_return,
        "sharpe": sharpe,
        "max_drawdown": max_dd,
    }

# print results
for k, v in results.items():
    print(f"\n===== {k} =====")
    for metric, val in v.items():
        print(f"{metric}: {val}")
```

```
===== 20bps =====
total_return: 0.07537821726153493
sharpe: 0.770861523873973
max_drawdown: -0.20755157649101985

===== 30bps =====
total_return: 0.008696393296915739
sharpe: 0.3772459235600509
max_drawdown: -0.21901945559788294

===== 40bps =====
total_return: -0.05391138310609267
sharpe: -0.016369676753871923
max_drawdown: -0.23545836215610827
```

1.6.26 25. Reduce trades (high-quality subset)

[280]:
```
### keep only strongest condition
reduced_df = trade_df[trade_df["use_lag2"]].copy()

print("original trades:", len(trade_df))
print("reduced trades:", len(reduced_df))
print("reduction %:", 1 - len(reduced_df)/len(trade_df))

REDUCED_DF_FILE = STRATEGY_DIR / "reduced_df.pkl"
reduced_df.to_pickle(REDUCED_DF_FILE)
```

```
manifest[stage_label]["reduced_df"] = str(REDUCED_DF_FILE)
print("saved:", REDUCED_DF_FILE)
print("shape:", reduced_df.shape)
```

```
original trades: 1470
reduced trades: 547
reduction %: 0.6278911564625851
saved: ../output/004_strategy/reduced_df.pkl
shape: (547, 16)
```

1.6.27 26. PnL for reduced trades

[281]:
```
daily_returns = (
    reduced_df
    .groupby("event_date")["signal_return"]
    .mean()
    .sort_index()
)

equity_curve = (1 + daily_returns).cumprod()

total_return = equity_curve.iloc[-1] - 1
volatility = daily_returns.std() * np.sqrt(trading_days)

sharpe = (
    daily_returns.mean() /
    daily_returns.std()
) * np.sqrt(trading_days)

drawdown = equity_curve / equity_curve.cummax() - 1
max_drawdown = drawdown.min()

print("\n===== REDUCED PERFORMANCE =====")
print("total_return:", total_return)
print("sharpe:", sharpe)
print("max_drawdown:", max_drawdown)
```

```
===== REDUCED PERFORMANCE =====
total_return: 0.375825052670274
sharpe: 3.81737605989996
max_drawdown: -0.11981425265943557
```

1.6.28 27. Calculate stress tests for reduced trades

[282]:
```
cost_levels = {
    "20bps": 0.002,
    "30bps": 0.003,
    "40bps": 0.004,
}

results = {}

for label, cost in cost_levels.items():

    df_cost = reduced_df.copy()

    # subtract cost per trade
    df_cost["net_return"] = df_cost["signal_return"] - cost

    # aggregate daily
    daily = (
        df_cost
        .groupby("event_date")["net_return"]
        .mean()
        .sort_index()
    )

    # equity
    equity = (1 + daily).cumprod()

    # stats
    total_return = equity.iloc[-1] - 1
    vol = daily.std() * np.sqrt(trading_days)

    sharpe = (
        daily.mean() /
        daily.std()
    ) * np.sqrt(trading_days)

    dd = equity / equity.cummax() - 1
    max_dd = dd.min()

    results[label] = {
        "total_return": total_return,
        "sharpe": sharpe,
        "max_drawdown": max_dd,
    }

# print results
for k, v in results.items():
```

```
    print(f"\n===== {k} =====")
    for metric, val in v.items():
        print(f"{metric}: {val}")
```

```
===== 20bps =====
total_return: 0.2732343749792987
sharpe: 2.9575792048840595
max_drawdown: -0.12880842980067475

===== 30bps =====
total_return: 0.22477350177937638
sharpe: 2.527680777376111
max_drawdown: -0.13327786403218322

===== 40bps =====
total_return: 0.17811151432890582
sharpe: 2.0977823498681607
max_drawdown: -0.1377289377277774
```

1.6.29 28. Pair Hit Ratio

[283]:
```
pair_hit = (
    signal_df
    .groupby(["reference_coin", "target_coin"])["correct_direction"]
    .mean()
    .sort_values(ascending=False)
)

pair_hit.head(10)

PAIR_HIT_FILE = STRATEGY_DIR / "pair_hit.pkl"
pair_hit.to_pickle(PAIR_HIT_FILE)
manifest[stage_label]["pair_hit"] = str(PAIR_HIT_FILE)
print("saved:", PAIR_HIT_FILE)
print("shape:", pair_hit.shape)
```

```
saved: ../output/004_strategy/pair_hit.pkl
shape: (72,)
```

1.6.30 29. Target Coin Direction and Count Tally

[284]:
```
pair_counts = (
    signal_df
    .groupby(["reference_coin", "target_coin"])
    .size()
)
```

```
top_pairs = pair_hit.head(10)

pd.concat([top_pairs, pair_counts], axis=1).head(10)
```

[284]:

		correct_direction	0
reference_coin	target_coin		
LTCUSDT	DOGEUSDT	0.684211	19
	AVAXUSDT	0.684211	19
XRPUSDT	LTCUSDT	0.666667	18
SOLUSDT	DOGEUSDT	0.666667	18
	BNBUSDT	0.666667	18
ETHUSDT	BNBUSDT	0.636364	22
	DOGEUSDT	0.636364	22
LTCUSDT	ETHUSDT	0.631579	19
	ADAUSDT	0.631579	19
	BNBUSDT	0.631579	19

1.6.31 30. Stress Test Pairs for high counts.

[285]:
```
min_count = 20

valid_pairs = pair_counts[pair_counts >= min_count].index

filtered_df = signal_df.set_index(
    ["reference_coin", "target_coin"]
).loc[valid_pairs].reset_index()

filtered_df["correct_direction"].mean()
```

[285]: np.float64(0.5093333333333333)

1.6.32 999. Persist Output

[286]:
```
manifest[stage_label]["timestamp"] = datetime.now(UTC).isoformat()
pd.to_pickle(manifest, MANIFEST_FILE)
print("manifest saved:", MANIFEST_FILE)
print("pipeline stages:", sorted(manifest.keys()))
print(f"artifacts in {stage_label}:", sorted(manifest[stage_label].keys()))
```

```
manifest saved: ../output/manifest.pkl
pipeline stages: ['001_download', '002_enrich', '003_analysis', '004_strategy']
artifacts in 004_strategy: ['daily_returns', 'event_response_df',
'event_response_expanded', 'pair_hit', 'reduced_df', 'timestamp', 'trade_df']
```

[ ]: